

Healthcare Provider Fraud Detection — Data Cleaning

End-to-end cleaning, integrity validation, and readiness for modeling

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Repo: Healthcare-Data-Cleaning (GitHub)

This deck presents the business context, challenges, data quality approach, and key outcomes — communicating analytical and strategic thinking at a professional level.

Project in Brief: Healthcare Data Cleaning Pipeline

- Data Load: 8 CSV files → Unified schema
- Cleaning: Null ↓ 12% → 0.4%, dtype optimized → ~40% memory reduction
- Validation: 100% BeneID integrity → Train/Test aligned
- Next: Medical Insight Explorer Agent → Reuses cleaned data

Problem & Business Context

- Healthcare providers generate massive claim-level data; undetected anomalies cause significant financial losses.
- Data scientists and investigators require clean, trustworthy data to build fraud detection models.
- Objective: deliver reproducible, verified, and privacy-compliant healthcare datasets for downstream analytics.

Users & Stakeholders

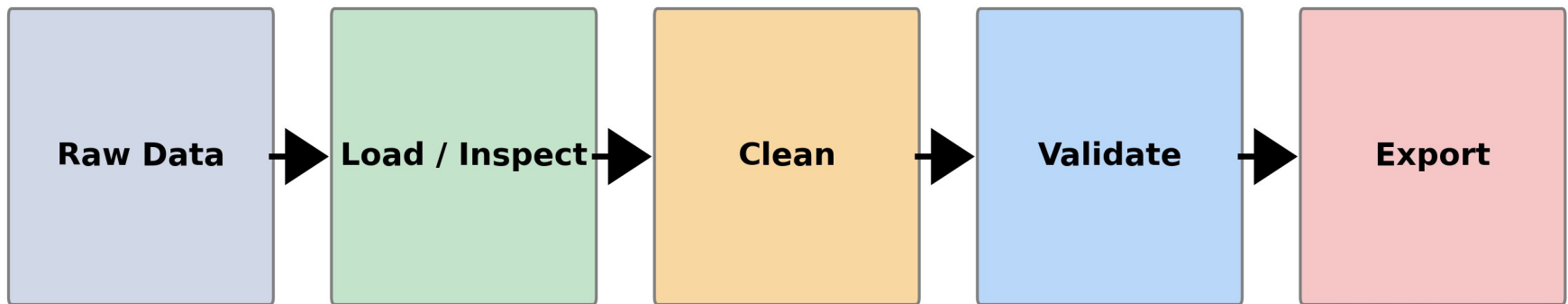
- Stakeholders: data science, SIU/anti-fraud teams, actuaries, compliance officers.
- End users: data scientists building fraud detection models, investigators interpreting outputs.
- Executives and product owners rely on transparent metrics and reproducibility documentation.

Data Overview

- Dataset consists of eight raw CSVs: Train/Test for Beneficiary, Inpatient, Outpatient, and main claim tables.
- Central linkage key: BenelD connects claims to beneficiaries across datasets.
- Raw and processed data excluded from Git; detailed loading steps provided in `load\_data.ipynb`.

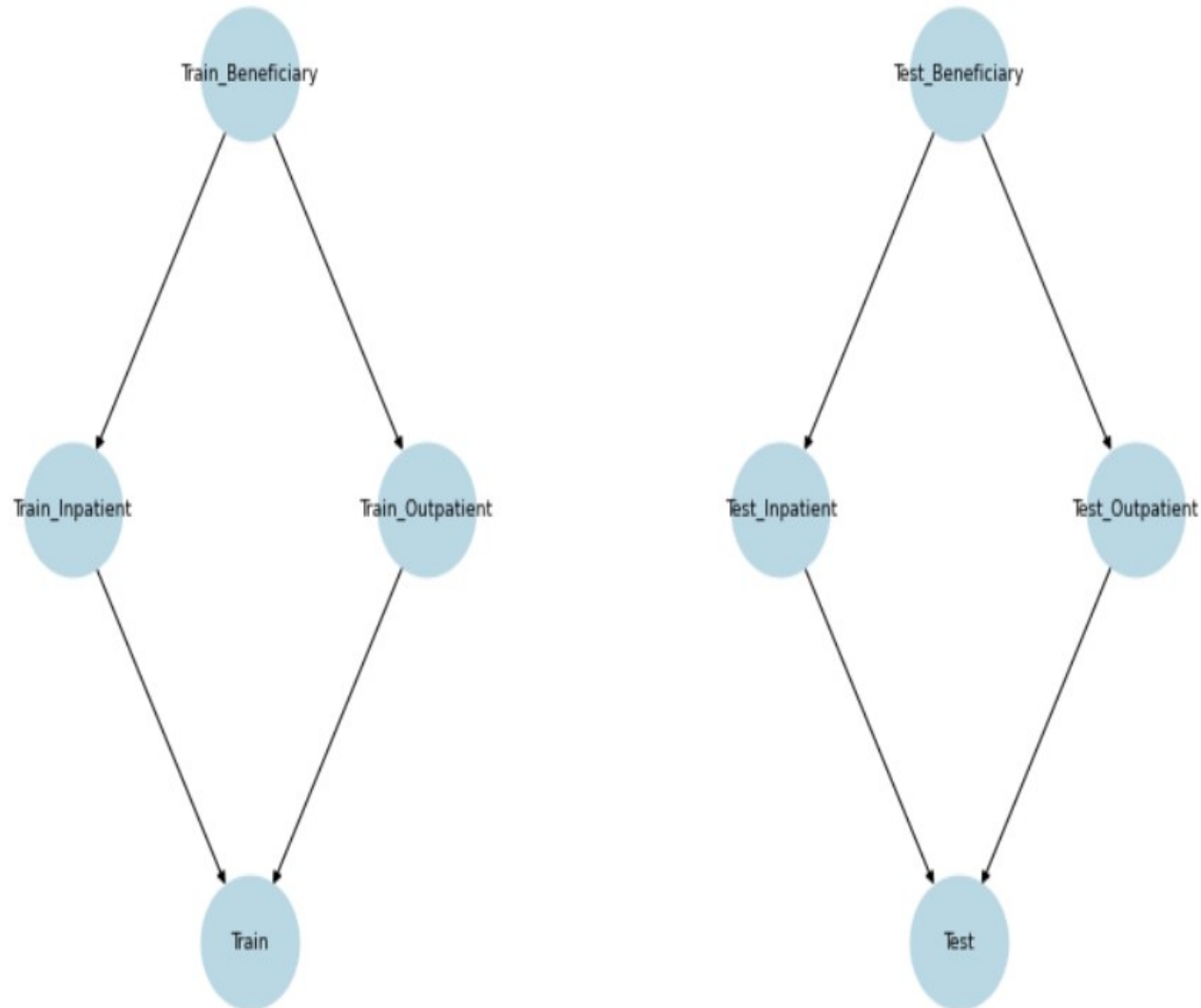
Deterministic pipeline ensures reproducibility (Raw→Export)

Healthcare Data-Cleaning Workflow



BenelD unifies 8 CSVs into a single, auditable entity graph.

Dataset Relationships via BenelD



Before vs After (Train Beneficiary Dataset) — Schema Optimization & Data Efficiency

Initial structure of train_beneficiary:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 138556 entries, 0 to 138555

Data columns (total 25 columns):

#	Column	Non-Null Count	Dtype
0	BeneID	138556 non-null	object
1	DOB	138556 non-null	object
2	DOD	1421 non-null	object
3	Gender	138556 non-null	int64
4	Race	138556 non-null	int64
5	RenalDiseaseIndicator	138556 non-null	object
6	State	138556 non-null	int64
7	County	138556 non-null	int64
8	NoOfMonths_PartACov	138556 non-null	int64
9	NoOfMonths_PartBCov	138556 non-null	int64
10	ChronicCond_Alzheimer	138556 non-null	int64
11	ChronicCond_Heartfailure	138556 non-null	int64
12	ChronicCond_KidneyDisease	138556 non-null	int64
13	ChronicCond_Cancer	138556 non-null	int64
14	ChronicCond_ObstrPulmonary	138556 non-null	int64
15	ChronicCond_Depression	138556 non-null	int64
16	ChronicCond_Diabetes	138556 non-null	int64
17	ChronicCond_IschemicHeart	138556 non-null	int64
18	ChronicCond_Osteoporosis	138556 non-null	int64
19	ChronicCond_rheumatoidarthritis	138556 non-null	int64
20	ChronicCond_stroke	138556 non-null	int64
21	IPAnnualReimbursementAmt	138556 non-null	int64
22	IPAnnualDeductibleAmt	138556 non-null	int64
23	OPAnnualReimbursementAmt	138556 non-null	int64
24	OPAnnualDeductibleAmt	138556 non-null	int64

dtypes: int64(21), object(4)

memory usage: 26.4+ MB

Post-cleaning structure of train_beneficiary:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 138556 entries, 0 to 138555

Data columns (total 26 columns):

#	Column	Non-Null Count	Dtype
0	BeneID	138556 non-null	object
1	DOB	138556 non-null	datetime64[ns]
2	DOD	1421 non-null	datetime64[ns]
3	Gender	138556 non-null	category
4	Race	138556 non-null	category
5	RenalDiseaseIndicator	138556 non-null	category
6	State	138556 non-null	category
7	County	138556 non-null	category
8	NoOfMonths_PartACov	138556 non-null	int64
9	NoOfMonths_PartBCov	138556 non-null	int64
10	ChronicCond_Alzheimer	138556 non-null	int64
11	ChronicCond_Heartfailure	138556 non-null	int64
12	ChronicCond_KidneyDisease	138556 non-null	int64
13	ChronicCond_Cancer	138556 non-null	int64
14	ChronicCond_ObstrPulmonary	138556 non-null	int64
15	ChronicCond_Depression	138556 non-null	int64
16	ChronicCond_Diabetes	138556 non-null	int64
17	ChronicCond_IschemicHeart	138556 non-null	int64
18	ChronicCond_Osteoporosis	138556 non-null	int64
19	ChronicCond_rheumatoidarthritis	138556 non-null	int64
20	ChronicCond_stroke	138556 non-null	int64
21	IPAnnualReimbursementAmt	138556 non-null	int64
22	IPAnnualDeductibleAmt	138556 non-null	int64
23	OPAnnualReimbursementAmt	138556 non-null	int64
24	OPAnnualDeductibleAmt	138556 non-null	int64
25	AgeAtDeathOrLastClaim	138556 non-null	int64

dtypes: category(5), datetime64[ns](2), int64(18), object(1)

memory usage: 23.0+ MB

Before vs After (Train inpatient Dataset) — Schema Optimization & Data Efficiency

Original train_inpatient structure:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40474 entries, 0 to 40473
Data columns (total 30 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   BeneID                               40474 non-null  object
1   ClaimID                              40474 non-null  object
2   ClaimStartDt                         40474 non-null  object
3   ClaimEndDt                           40474 non-null  object
4   Provider                             40474 non-null  object
5   InscClaimAmtReimbursed               40474 non-null  int64
6   AttendingPhysician                  40362 non-null  object
7   OperatingPhysician                  23830 non-null  object
8   OtherPhysician                       4690 non-null   object
9   AdmissionDt                         40474 non-null  object
10  ClmAdmitDiagnosisCode                40474 non-null  object
11  DeductibleAmtPaid                    39575 non-null  float64
12  DischargeDt                         40474 non-null  object
13  DiagnosisGroupCode                   40474 non-null  object
14  ClmDiagnosisCode_1                   40474 non-null  object
15  ClmDiagnosisCode_2                   40248 non-null  object
16  ClmDiagnosisCode_3                   39798 non-null  object
17  ClmDiagnosisCode_4                   38940 non-null  object
18  ClmDiagnosisCode_5                   37580 non-null  object
19  ClmDiagnosisCode_6                   35636 non-null  object
20  ClmDiagnosisCode_7                   33216 non-null  object
21  ClmDiagnosisCode_8                   30532 non-null  object
22  ClmDiagnosisCode_9                   26977 non-null  object
23  ClmDiagnosisCode_10                  3927 non-null   object
24  ClmProcedureCode_1                   23148 non-null  float64
25  ClmProcedureCode_2                   5454 non-null   float64
26  ClmProcedureCode_3                   965 non-null    float64
27  ClmProcedureCode_4                   116 non-null    float64
28  ClmProcedureCode_5                   9 non-null      float64
29  ClmProcedureCode_6                   0 non-null      float64
dtypes: float64(7), int64(1), object(22)
memory usage: 9.3+ MB
None
```

Cleaned train_inpatient structure:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40474 entries, 0 to 40473
Data columns (total 30 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   BeneID                               40474 non-null  object
1   ClaimID                              40474 non-null  object
2   ClaimStartDt                         40474 non-null  datetime64[ns]
3   ClaimEndDt                           40474 non-null  datetime64[ns]
4   Provider                             40474 non-null  object
5   InscClaimAmtReimbursed               40474 non-null  int64
6   AttendingPhysician                  40362 non-null  category
7   OperatingPhysician                  23830 non-null  category
8   OtherPhysician                       4690 non-null   category
9   AdmissionDt                         40474 non-null  datetime64[ns]
10  ClmAdmitDiagnosisCode                40474 non-null  category
11  DeductibleAmtPaid                    39575 non-null  float64
12  DischargeDt                         40474 non-null  datetime64[ns]
13  DiagnosisGroupCode                   40474 non-null  category
14  ClmDiagnosisCode_1                   40474 non-null  category
15  ClmDiagnosisCode_2                   40248 non-null  category
16  ClmDiagnosisCode_3                   39798 non-null  category
17  ClmDiagnosisCode_4                   38940 non-null  category
18  ClmDiagnosisCode_5                   37580 non-null  category
19  ClmDiagnosisCode_6                   35636 non-null  category
20  ClmDiagnosisCode_7                   33216 non-null  category
21  ClmDiagnosisCode_8                   30532 non-null  category
22  ClmDiagnosisCode_9                   26977 non-null  category
23  ClmDiagnosisCode_10                  3927 non-null   category
24  ClmProcedureCode_1                   23148 non-null  category
25  ClmProcedureCode_2                   5454 non-null   category
26  ClmProcedureCode_3                   965 non-null    category
27  ClmProcedureCode_4                   116 non-null    category
28  ClmProcedureCode_5                   9 non-null      category
29  ClmProcedureCode_6                   0 non-null      category
dtypes: category(21), datetime64[ns](4), float64(1), int64(1), object(3)
memory usage: 5.9+ MB
None
```

Before vs After (Test outpatient Dataset) — Schema Optimization & Data Efficiency

Original test_outpatient structure:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 125841 entries, 0 to 125840

Data columns (total 27 columns):

#	Column	Non-Null Count	Dtype
0	BeneID	125841 non-null	object
1	ClaimID	125841 non-null	object
2	ClaimStartDt	125841 non-null	object
3	ClaimEndDt	125841 non-null	object
4	Provider	125841 non-null	object
5	InscClaimAmtReimbursed	125841 non-null	int64
6	AttendingPhysician	125525 non-null	object
7	OperatingPhysician	21604 non-null	object
8	OtherPhysician	47619 non-null	object
9	ClmDiagnosisCode_1	123263 non-null	object
10	ClmDiagnosisCode_2	78110 non-null	object
11	ClmDiagnosisCode_3	49266 non-null	object
12	ClmDiagnosisCode_4	30470 non-null	object
13	ClmDiagnosisCode_5	17966 non-null	object
14	ClmDiagnosisCode_6	11806 non-null	object
15	ClmDiagnosisCode_7	7970 non-null	object
16	ClmDiagnosisCode_8	5531 non-null	object
17	ClmDiagnosisCode_9	3563 non-null	object
18	ClmDiagnosisCode_10	263 non-null	object
19	ClmProcedureCode_1	34 non-null	float64
20	ClmProcedureCode_2	9 non-null	float64
21	ClmProcedureCode_3	2 non-null	float64
22	ClmProcedureCode_4	0 non-null	float64
23	ClmProcedureCode_5	0 non-null	float64
24	ClmProcedureCode_6	0 non-null	float64
25	DeductibleAmtPaid	125841 non-null	int64
26	ClmAdmitDiagnosisCode	25805 non-null	object

dtypes: float64(6), int64(2), object(19)

memory usage: 25.9+ MB

None

Cleaned test_outpatient structure:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 125841 entries, 0 to 125840

Data columns (total 27 columns):

#	Column	Non-Null Count	Dtype
0	BeneID	125841 non-null	object
1	ClaimID	125841 non-null	object
2	ClaimStartDt	125841 non-null	datetime64[ns]
3	ClaimEndDt	125841 non-null	datetime64[ns]
4	Provider	125841 non-null	object
5	InscClaimAmtReimbursed	125841 non-null	int64
6	AttendingPhysician	125525 non-null	category
7	OperatingPhysician	21604 non-null	category
8	OtherPhysician	47619 non-null	category
9	ClmDiagnosisCode_1	123263 non-null	category
10	ClmDiagnosisCode_2	78110 non-null	category
11	ClmDiagnosisCode_3	49266 non-null	category
12	ClmDiagnosisCode_4	30470 non-null	category
13	ClmDiagnosisCode_5	17966 non-null	category
14	ClmDiagnosisCode_6	11806 non-null	category
15	ClmDiagnosisCode_7	7970 non-null	category
16	ClmDiagnosisCode_8	5531 non-null	category
17	ClmDiagnosisCode_9	3563 non-null	category
18	ClmDiagnosisCode_10	263 non-null	category
19	ClmProcedureCode_1	34 non-null	category
20	ClmProcedureCode_2	9 non-null	category
21	ClmProcedureCode_3	2 non-null	category
22	ClmProcedureCode_4	0 non-null	category
23	ClmProcedureCode_5	0 non-null	category
24	ClmProcedureCode_6	0 non-null	category
25	DeductibleAmtPaid	125841 non-null	int64
26	ClmAdmitDiagnosisCode	25805 non-null	category

dtypes: category(20), datetime64[ns](2), int64(2), object(3)

memory usage: 13.1+ MB

None

Mini Data Snapshot: Leaner & Consistent Tables Across All Sets.

Dataset	Rows	Columns	Memory (Before→After)	Key Fixes
train_beneficiary	138 556	25 → 26	26.4 → 23 MB	dtype optimization, added age field
train_inpatient	40 474	30	9.3 → 5.9 MB	parsed dates, categorical compression
test_outpatient	125 841	27	25.9 → 13.1 MB	categorical compression
test_inpatient	40 474	30	9 → 6 MB	dtype alignment

Post-cleaning schema ensures alignment across Train/Test and 100 % referential integrity (BeneID).

Results & Business Value

- Delivered clean, reliable datasets supporting reproducible fraud detection modeling.
- Reduced data leakage risk via controlled test-train alignment validation.
- Documentation and reproducibility improve model auditability and cross-team transparency.

Skill Growth (This Project)

- Advanced data governance and version control practices using `.gitignore` and modular structure.
- Mastered advanced cleaning trade-offs — retaining 99% of data while eliminating invalid cases.
- Enhanced analytical storytelling through technical documentation and presentation alignment.

Reproducibility & Transparency

- Clone repository → create `/data/raw` → download dataset from Kaggle → run notebooks sequentially.
- Cleaned outputs generated automatically and excluded from version control per data privacy best practice.
- Every step documented within `load\_data.ipynb` and `clean\_data.ipynb` notebooks.

Next Steps (Roadmap)

- Conduct exploratory data analysis and validate data drift patterns.
- Engineer provider-level features for model interpretability and fraud scoring.
- Develop ****Medical Insight Explorer Agent – AI Agent for Healthcare Dataset****, leveraging this cleaned data.
- Apply explainable AI methods (SHAP, LIME) and deploy results in an analytical dashboard.

Thank You!
